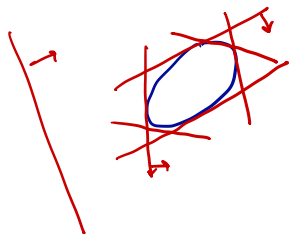


Geometric Concept of Duality

① $C \subseteq \mathbb{R}^n$ convex & closed

$\Leftrightarrow C = \bigcap_{H \supseteq C} H, \quad H \text{ half-space.}$



vs



② $x, C, x \notin C$



Duality and Optimality

$$\begin{aligned} \min: & f(x) \\ \text{st: } & x \in \mathcal{X} \end{aligned}$$

$$\hat{x} \text{ optimal} \quad \underline{\text{iff}} \quad \exists \hat{g} \in \partial f(\hat{x})$$
$$\underline{\text{and}} \quad \exists \hat{n} \in N_{\mathcal{X}}(\hat{x}) \quad \text{s.t.}$$

$$\boxed{0 = \hat{g} + \hat{n}} \quad \left(0 \in \partial f(\hat{x}) + N_{\mathcal{X}}(\hat{x}) \right)$$

$(\hat{n}, \hat{g})$ are certificates of optimality for $\hat{x}$.

These are the dual variables.

See: KKT conditions.

Duality for LP via min-max

$$\begin{aligned} \min: & c^T x \\ \text{st: } & Ax \geq b \end{aligned} = \min_x : \max_{p \geq 0} : c^T x + p^T (b - Ax)$$

$$\begin{aligned} \max_{p \geq 0} : \min_x : & p^T b + (c^T - p^T A)x \\ & \geq \end{aligned}$$

Weak duality

$$\left. \begin{aligned} \max: & p^T b \\ \text{st: } & p^T A = c^T \\ & p \geq 0 \end{aligned} \right\} \text{ dual problem.}$$

Fact: If primal or dual is feasible with finite opt value, then both are, and $c^T x^* = p^{*T} b$.
 i.e. " $\geq$ " is " $=$ " (Strong Duality).

General LP Duality

Exercise : show

$$\begin{aligned} \text{(P)} \quad & \min: c^T x \\ & \text{st: } a_i^T x = b_i, \quad i \in M_1 \\ & \quad \quad a_i^T x \geq b_i, \quad i \in M_2 \\ & \quad \quad x_j \geq 0 \quad j \in N_1 \\ & \quad \quad x_j \text{ free} \quad j \in N_2 \end{aligned}$$

(D)

$$\begin{aligned} \text{max: } & p^T b \\ \text{st: } & p_i \text{ free } i \in M_1 \\ & p_i \geq 0 \quad i \in M_2 \\ & p^T A_j \leq c_j \quad j \in N_1 \\ & p^T A_j = c_j, \quad j \in N_2 \end{aligned} \quad \left| \right|$$

Exercise

Back to slide 2: write the normal cone at a vertex. What is the connection to the dual variables?

$$\begin{array}{ll} \min: & f(x) \\ \text{st:} & x \in \mathcal{X} \end{array}$$

$$\hat{x} \text{ opt} \quad \text{iff}$$

$$0 \in \partial f(\hat{x}) + \underbrace{N_{\mathcal{X}}(\hat{x})}_{\text{normal cone}}$$

Pick a simple const. opt:

min: quad

st: 2 quad const.

$$(x_1 - 3)^2 + 2(x_2 - 2)^2$$

